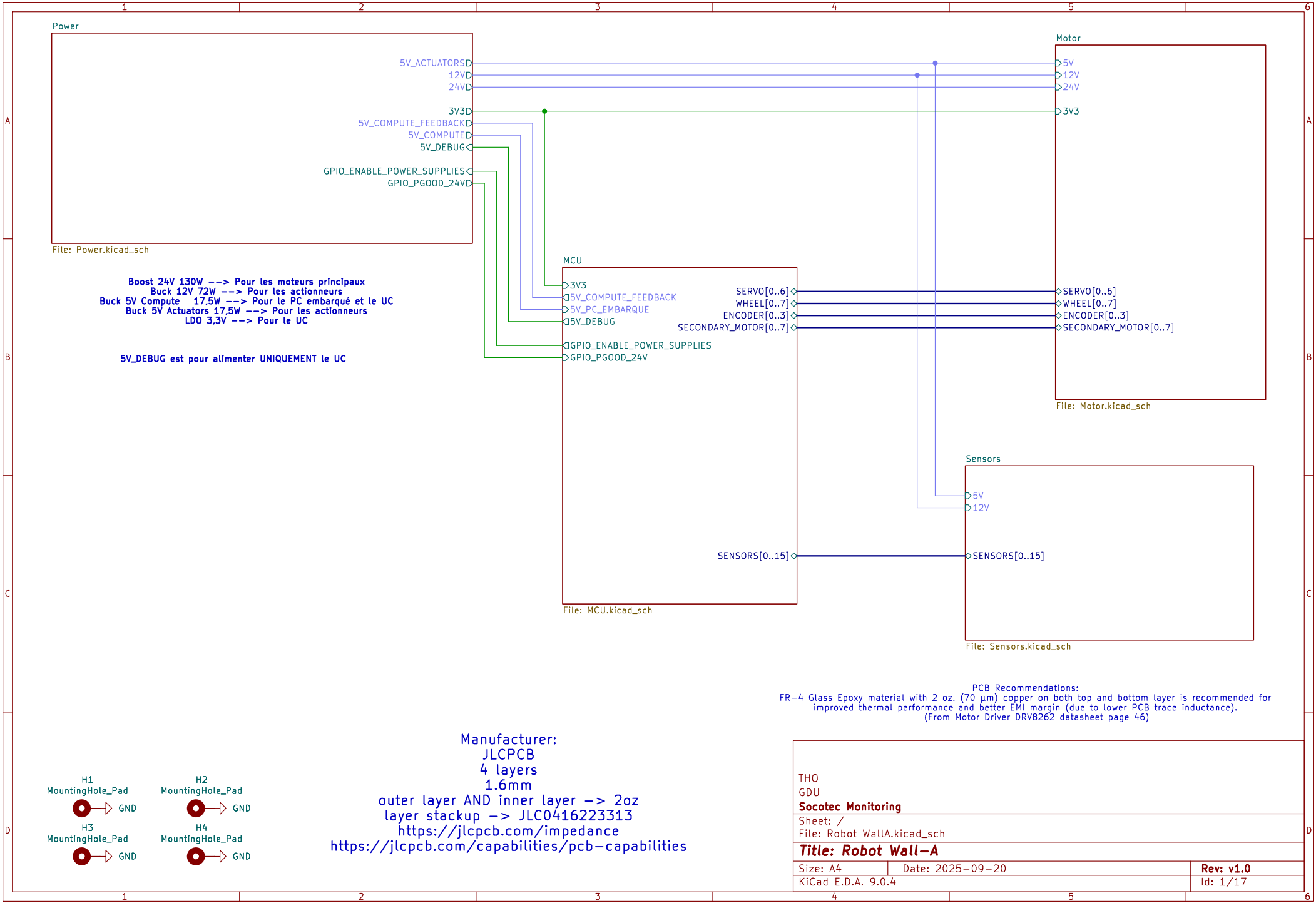
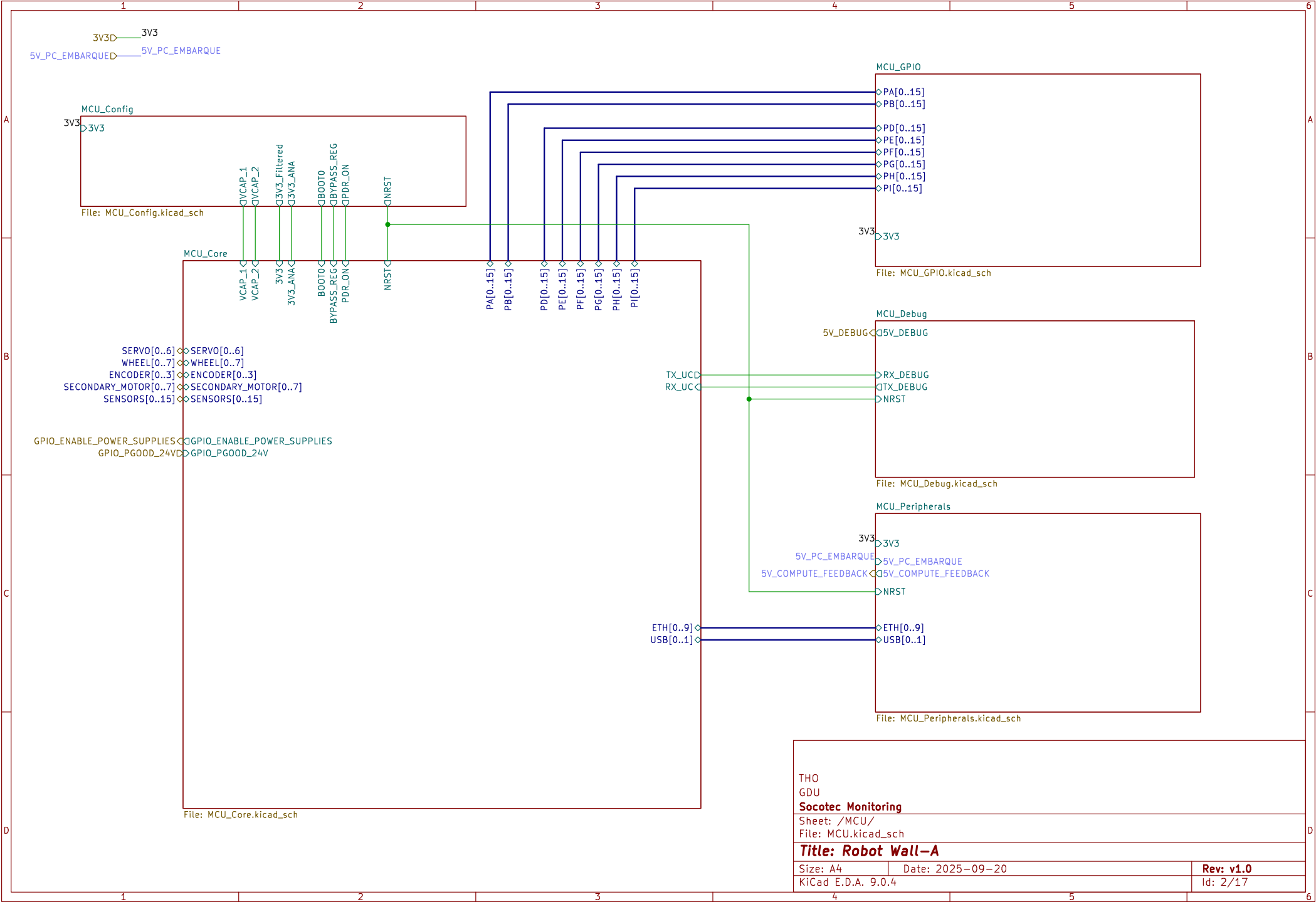






TH0
GDU

Socotec Monitoring

Sheet: /MCU/
File: MCU.kicad_sch

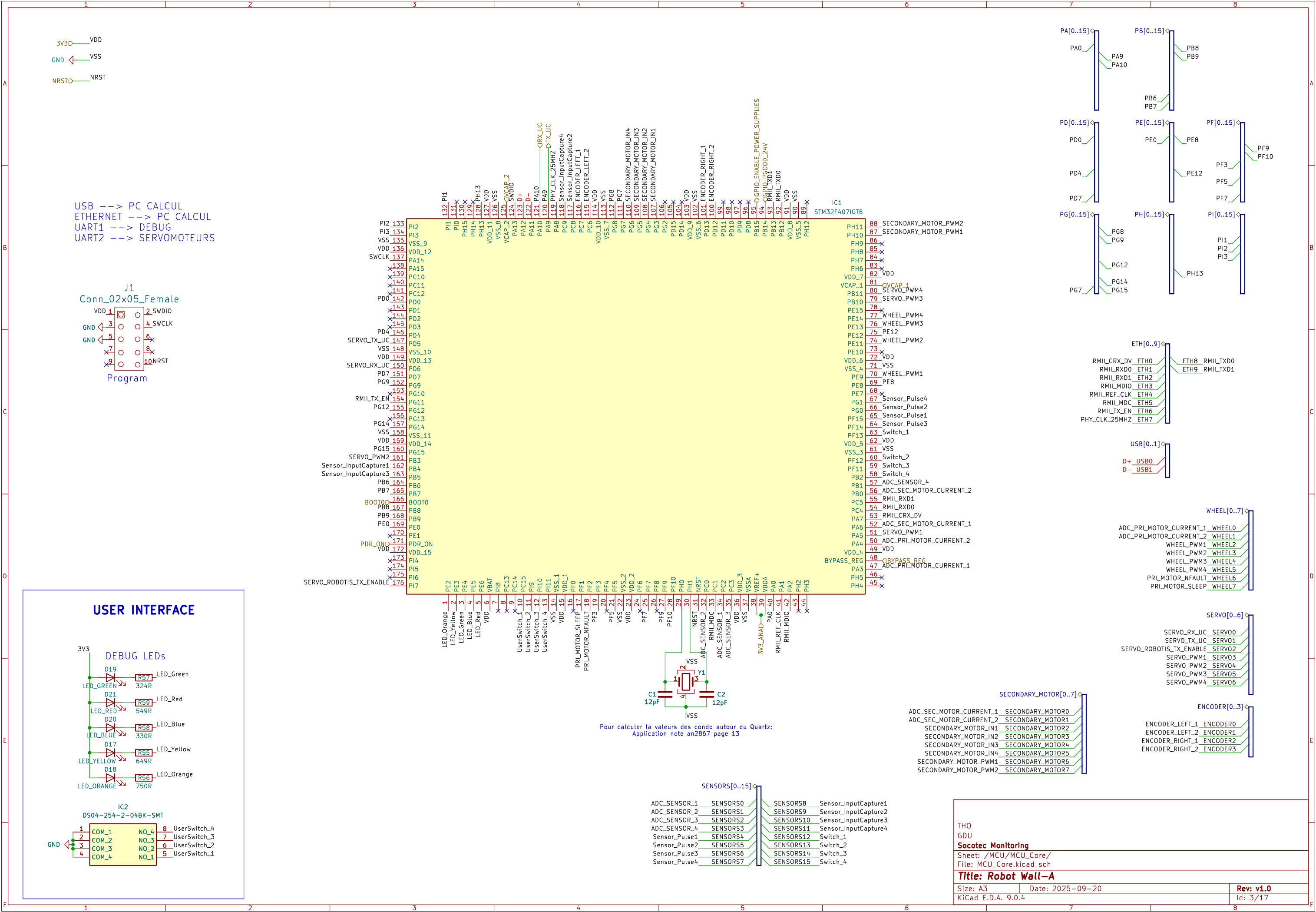
Title: Robot Wall-A

Size: A4 Date: 2025-09-20

KiCad E.D.A. 9.0.4

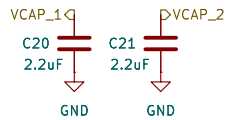
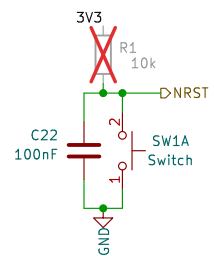
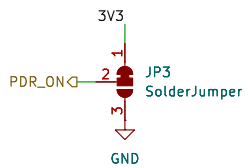
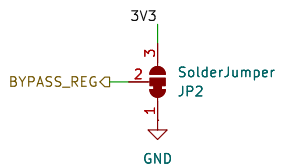
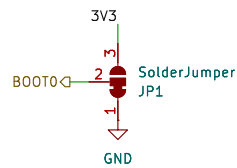
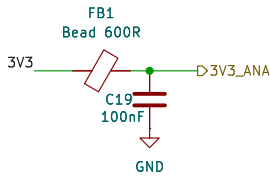
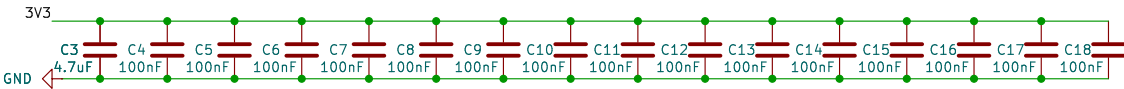
Rev: v1.0

Id: 2/17



Must be as close as possible from μC

3V3 $\rightarrow$ 3V3_Filtered

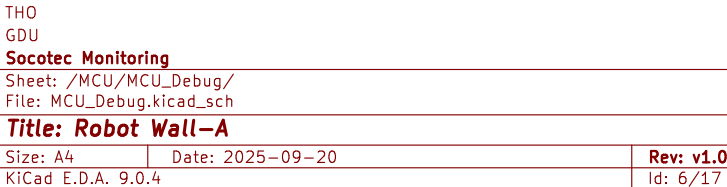


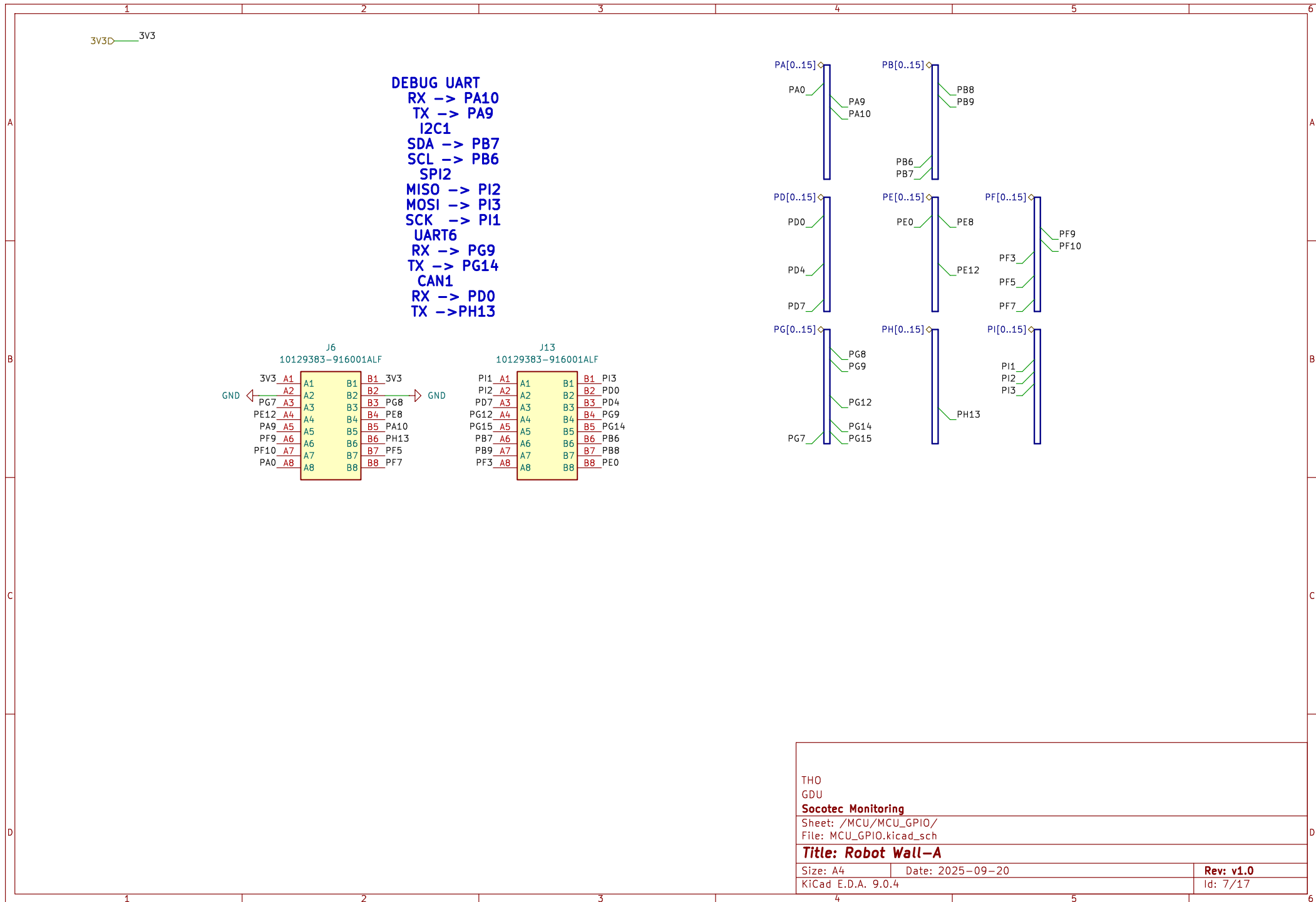
TH0		
GDU		
Socotec Monitoring		
Sheet: /MCU/MCU_Config/		
File: MCU_Config.kicad_sch		
Title: Robot Wall-A		
Size: A4	Date: 2025-09-20	Rev: v1.0
KiCad E.D.A. 9.0.4	Id: 4/17	

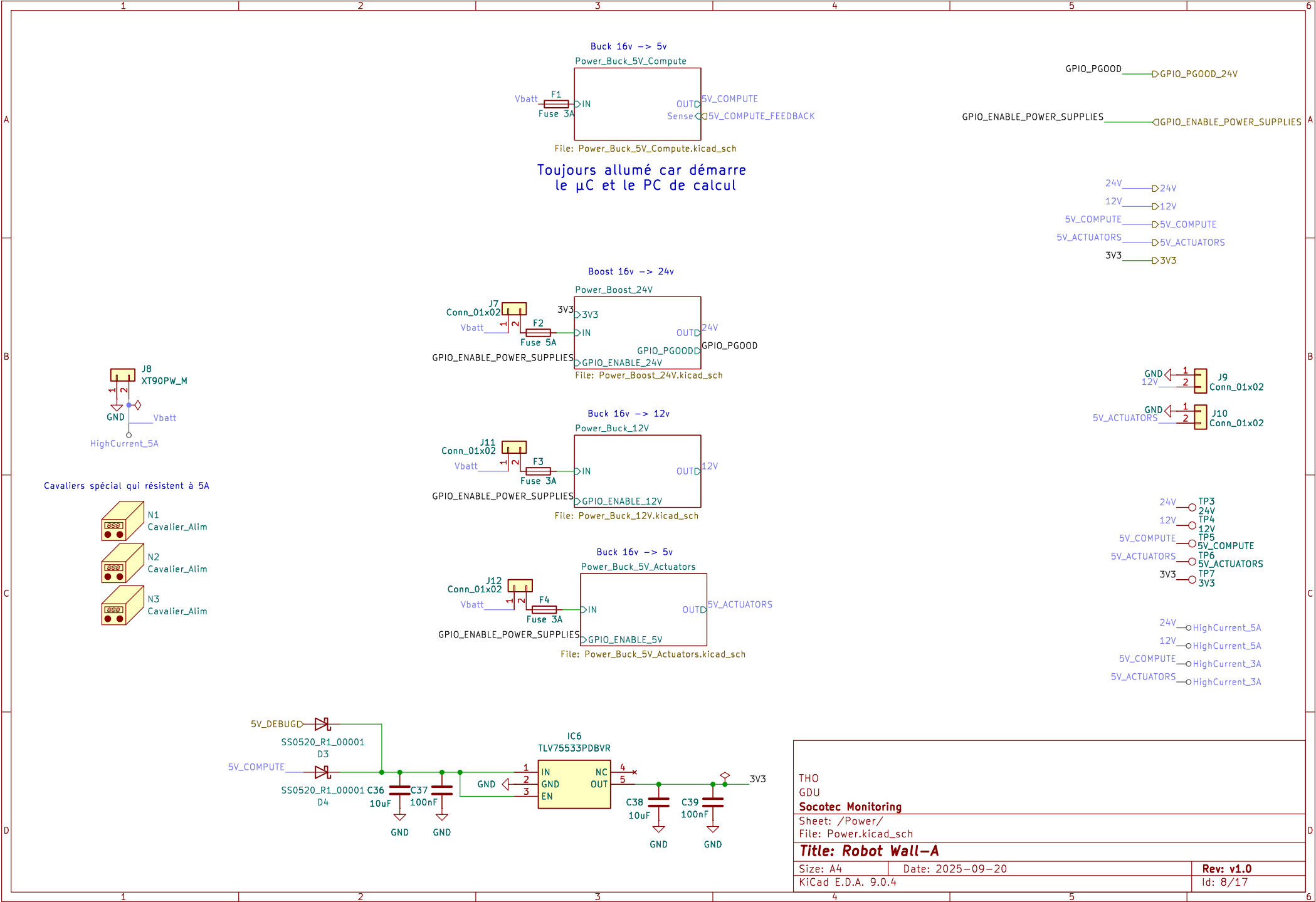
[illegible]

Id: 5/17

<https://www.youtube.com/watch?v=M13QxtPVrXY&t=798s>

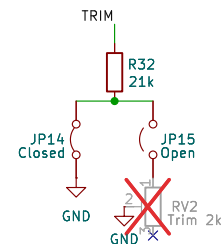
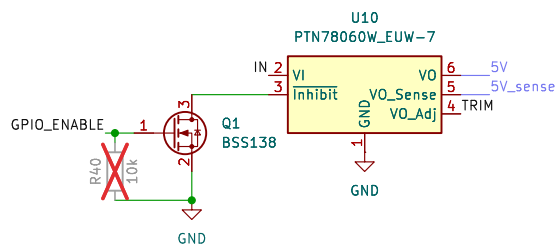
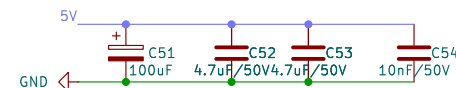
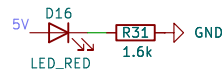
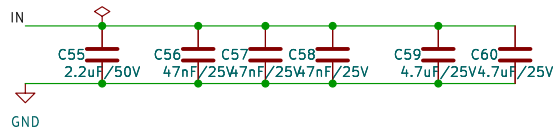






IND IN

5V OUT
5V_sense Sense



TH0
GDU

Socotec Monitoring

Sheet: /Power/Power_Buck_5V_Compute/
File: Power_Buck_5V_Compute.kicad_sch

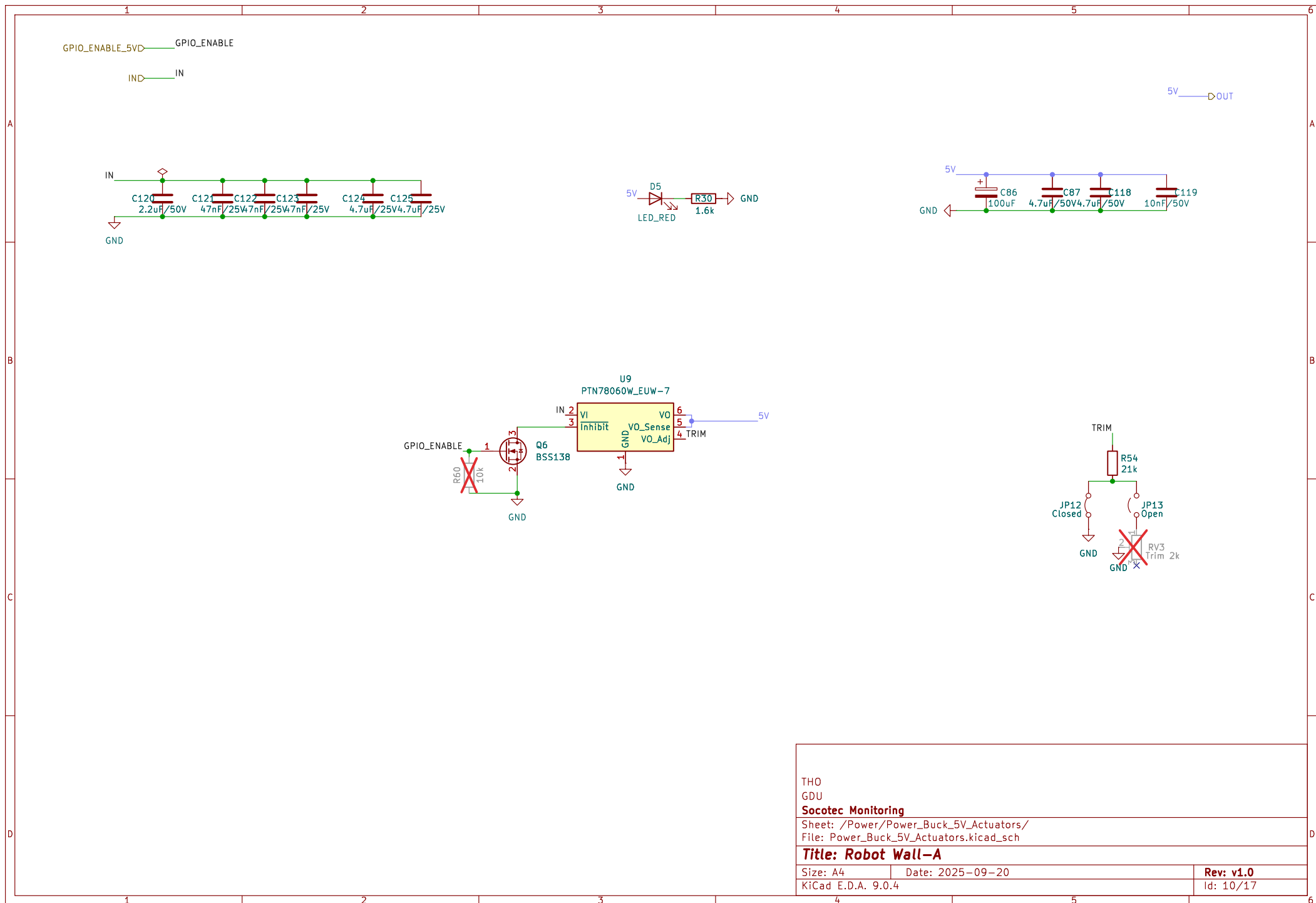
Title: Robot Wall-A

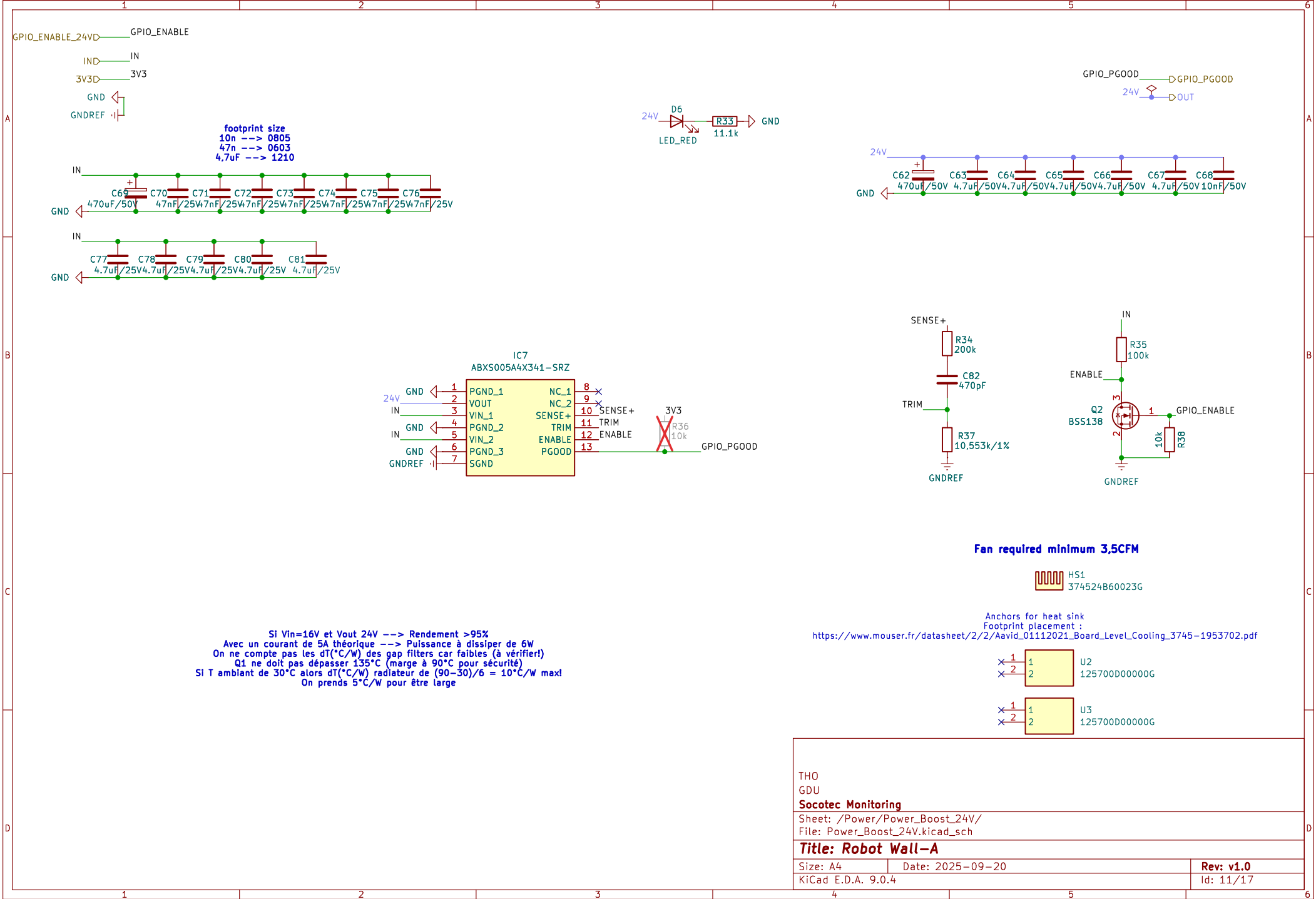
Size: A4 Date: 2025-09-20

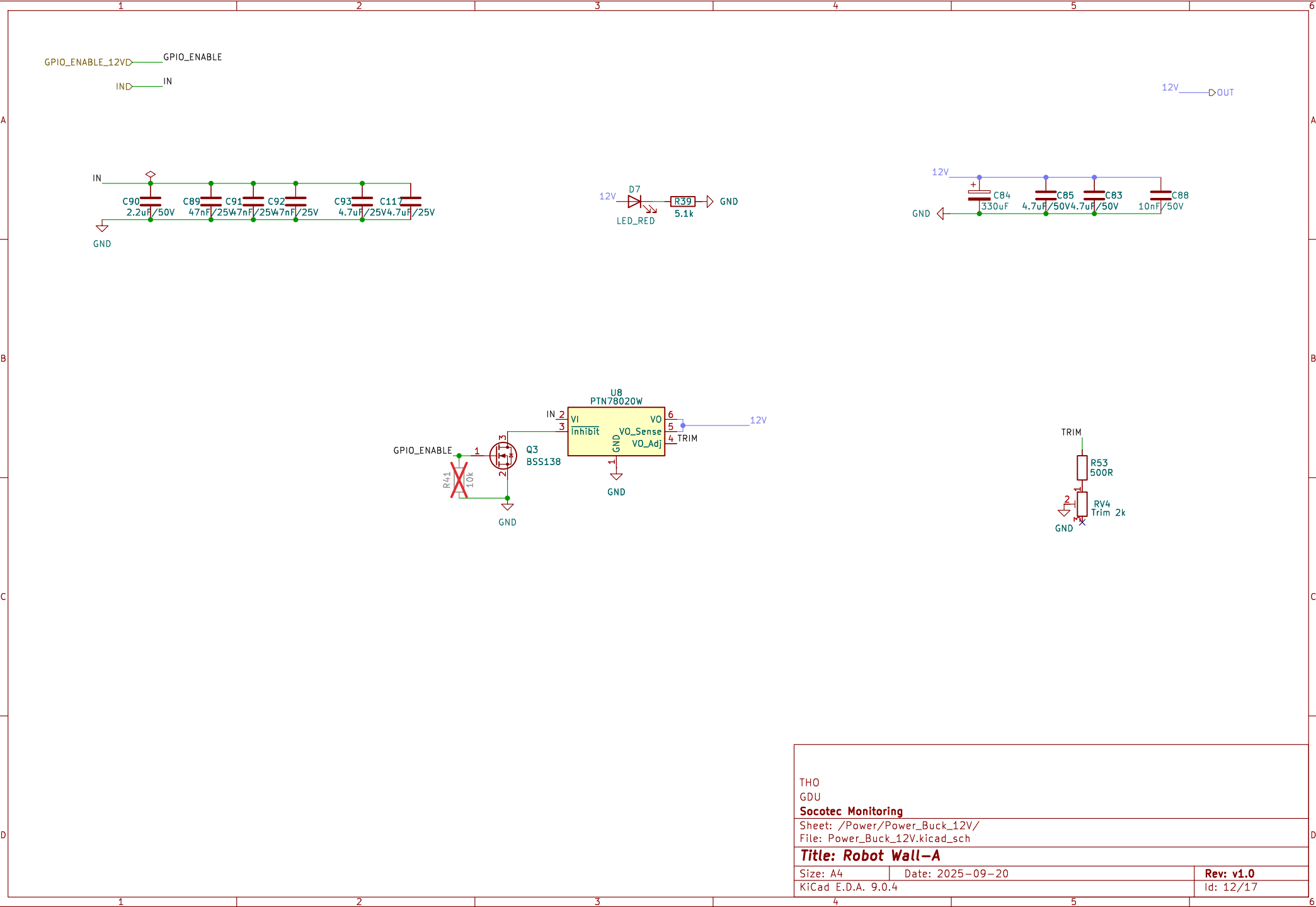
KiCad E.D.A. 9.0.4

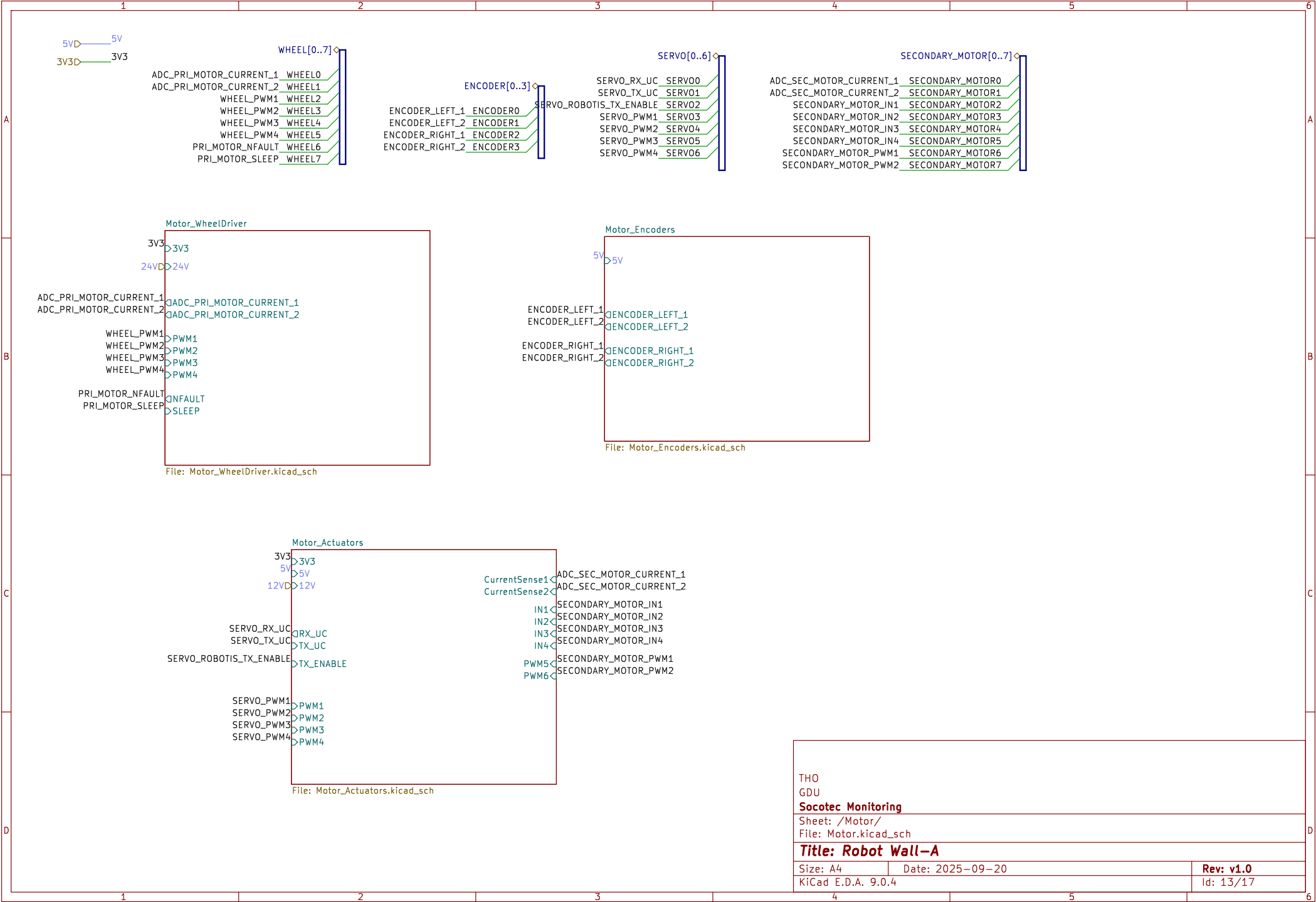
Rev: v1.0

Id: 9/17







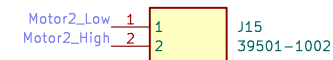
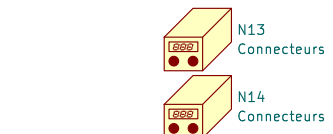
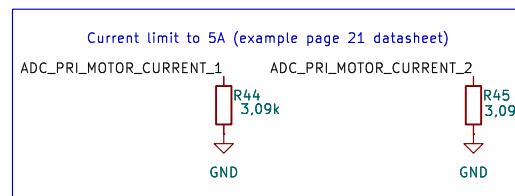
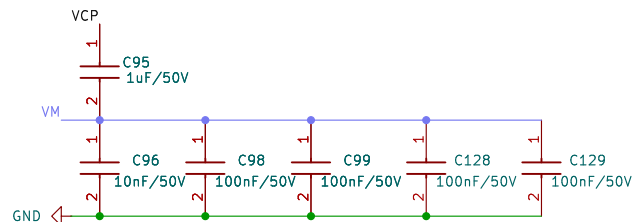
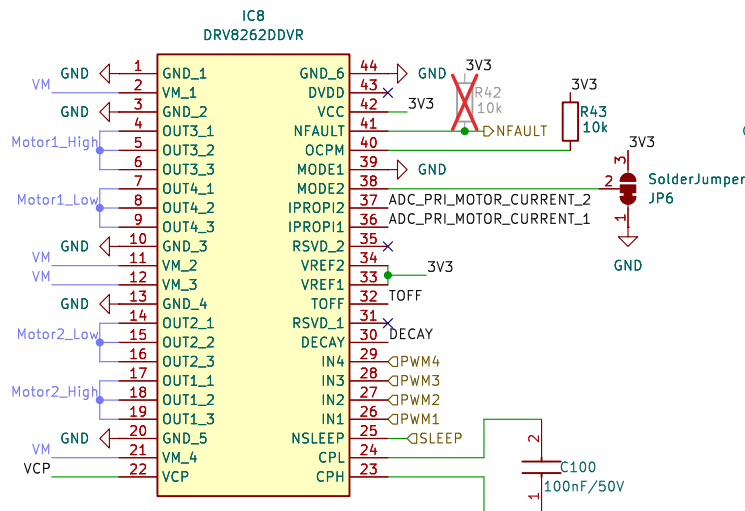


3V3D — 3V3
24VD — VM



Default Configuration:
OCPM = 1 --> Over Current Protection error = Retry
MODE = 0 --> PH/EN mode
Toff = 1 --> Waiting time between change = 16µs
Decay = HZ --> Auto detect mode

ADC_PRI_MOTOR_CURRENT_1 —> ADC_PRI_MOTOR_CURRENT_1
ADC_PRI_MOTOR_CURRENT_2 —> ADC_PRI_MOTOR_CURRENT_2



ATTENTION!!
Pour simplifier le routage,
les phases du moteurs on été inversées.
Attention au câblage!

https://www.mouser.fr/datasheet/2/2/Aavid_01112021_Board_LeveL_Cooling_3745-1953702.pdf

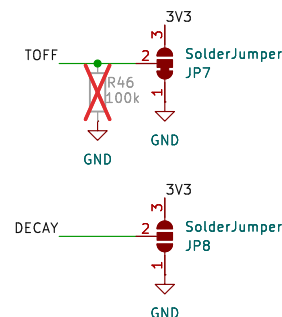


Fan required minimum 3,5CFM

2x Transmotec PD4266-24-17-BFEC
24V / 2,1A nominal
4,2A nominal total

Power loss calculation and dissipator estimation
(page 33 to 35 datasheet)
Calculations already done in datasheet (same example with 4A)
Conclusion: Dissipator with less than 4°C/W

Heat sink possibilities:
WAVE-35-12 with anchors 125700D00000G --> Fan required minimum 3CFM
374524B60023G with anchors 125700D00000G --> Fan required minimum 3,5CFM
567-45AB with screws --> Fan required minimum 4CFM



Motor1_High — HighCurrent_5A
Motor1_Low — HighCurrent_5A
Motor2_High — HighCurrent_5A
Motor2_Low — HighCurrent_5A

THO
GDU

Socotec Monitoring

Sheet: /Motor/Motor_WheelDriver/
File: Motor_WheelDriver.kicad_sch

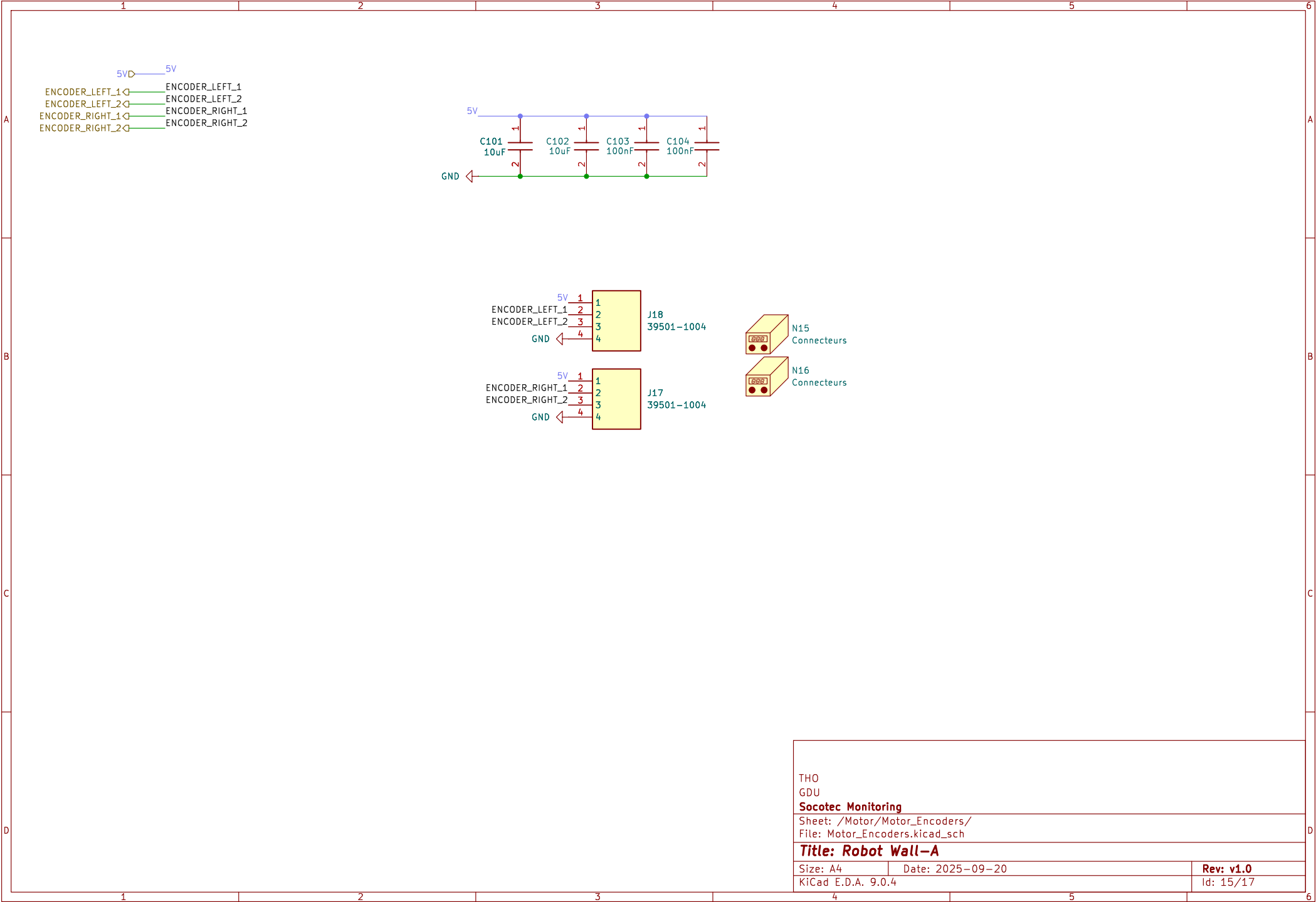
Title: Robot Wall-A

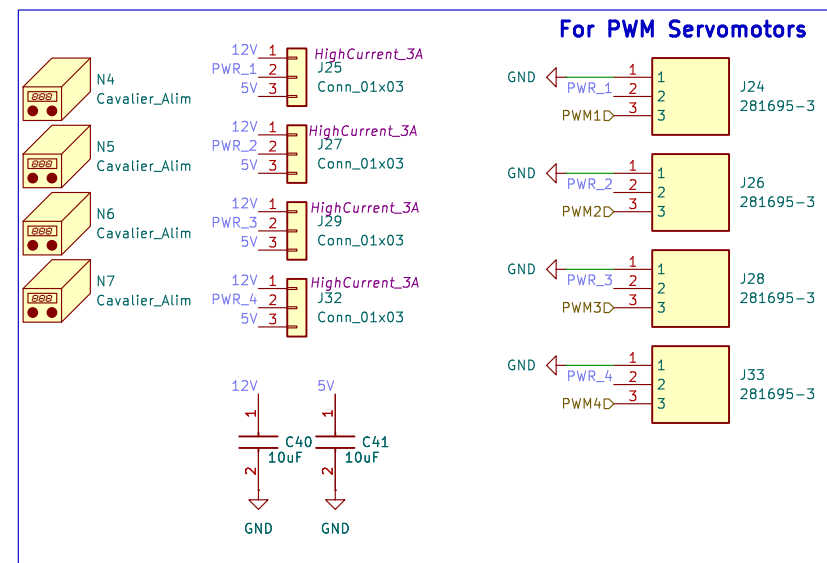
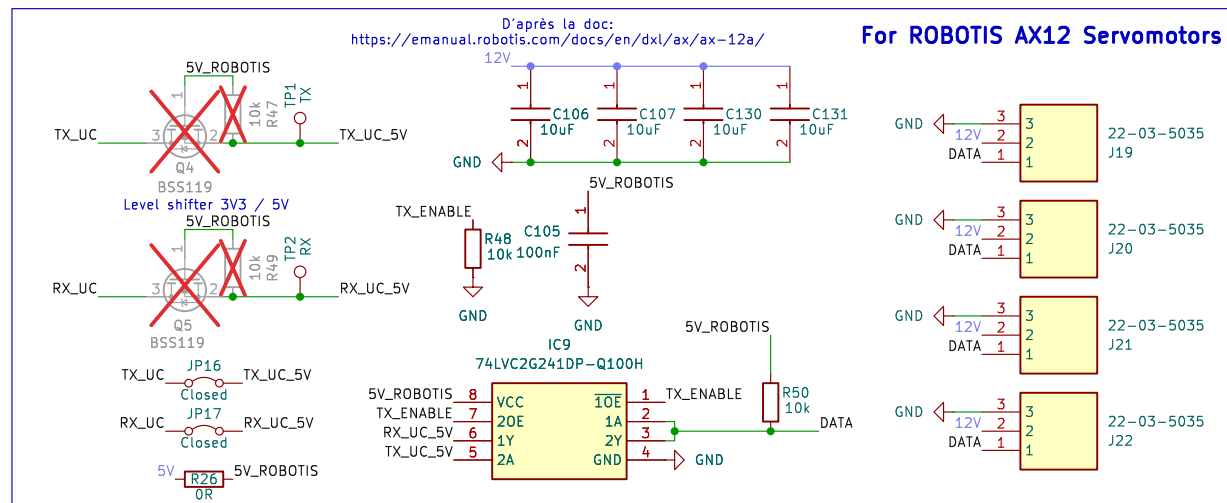
Size: A4 Date: 2025-09-20

KiCad E.D.A. 9.0.4

Rev: v1.0

Id: 14/17





Id: 16/17

